

moe-interp

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Abstract

Sparse Mixture-of-Experts (MoE) layers have become a standard way to grow language models, increasing their parameter counts while keeping compute required per token nearly constant. Many studies examine how these models route tokens and what exactly is computed by individual experts. These studies, however, use vastly different experimental setups (different models, different definitions of fundamental terms, etc.), making their results difficult to compare directly. For a significant fraction of these studies, public implementations have not been released. For the ones that have been released, the implementations are still often limited - they rely on a single model family or depend on a particular library version. Even fundamental quantities such as routing weight carry different meanings across router designs, making cross-model comparisons harder. We aim to develop *moe-interp*, a library that records routing scores, weights and expert activations in a common format across models, applies controlled interventions to routers and experts, and runs the analyses of recent studies on any supported model that provides the signals they require.

1 How do Mixture-of-Experts layers route tokens?

In a standard (dense) transformer, every token in a given layer passes through the same feed-forward network. In a Mixture-of-Experts model, this block is replaced by several *experts*. A router assigns each token to a small subset of these experts and determines how strongly each selected expert’s output is weighted.

Formally, an MoE layer replaces the feed-forward block of a transformer (Vaswani et al., 2017) with N expert networks and a router that assigns each token to a small subset of them (Shazeer et al., 2017; Lepikhin et al., 2021; Fedus et al., 2022). For the hidden state $\mathbf{h} \in \mathbb{R}^d$ of a single token, the layer computes

$$\begin{aligned} \mathbf{s} &= \sigma(\mathbf{W}_r \mathbf{h}) \\ \mathbf{y} &= \sum_{i \in \mathcal{T}_k(\mathbf{s})} g_i(\mathbf{s}) E_i(\mathbf{h}) + \sum_{j=1}^S E_j^{\text{sh}}(\mathbf{h}) \end{aligned} \tag{1}$$

where $\mathbf{W}_r \in \mathbb{R}^{N \times d}$ is the router’s weight matrix, σ is the scoring function of the router, $\mathcal{T}_k(\mathbf{s})$ is the set of the k experts with the largest scores, g_i is the weight applied to the output of expert E_i , and $E_1^{\text{sh}}, \dots, E_S^{\text{sh}}$ are shared experts that process every token (Dai et al., 2024).

Although all MoE models follow this basic pattern, they differ in how routing scores are computed, how experts are selected, and how the selected experts are weighted. Each component of eq. (1) varies across contemporary models, as the following examples illustrate:

- **Score function.** OLMoE (Muennighoff et al., 2025), Mixtral (Jiang et al., 2024) and Qwen3 (Yang et al., 2025) apply a softmax over all experts, DeepSeek-V3 (DeepSeek-AI, 2024b) applies a sigmoid, and geometric routers apply a softmax to the cosine similarity between a normalized projection of $\mathbf{h}$ and learned expert centroids (Ter-novtsii & Bilak, 2026a).

- **Order of selection and normalization.** gpt-oss (OpenAI, 2025) selects experts from the raw logits and applies the softmax to the selected experts only. Mixtral and Qwen3 renormalize the selected probabilities, whereas OLMoE uses them unchanged.
- **Selection rule.** DeepSeek-V3 adds a per-expert bias, adjusted during training to balance the load, to the scores used for selection, restricts the choice to a subset of expert groups, and multiplies the resulting weights by a constant.
- **Shared experts.** DeepSeek models (Dai et al., 2024; DeepSeek-AI, 2024a) add experts that are always active, and Qwen1.5-MoE scales its shared expert by a learned gate.

These differences create two problems for interpretability research:

- **Quantities do not transfer between models.** The routing weight is a probability over all experts in OLMoE, a probability over the selected experts in gpt-oss, and a rescaled sigmoid score in DeepSeek-V3. Statistics such as routing entropy or expert contribution computed from these tensors, are not comparable unless the differences are made explicit.
- **Analyses are tied to single models.** Each study is carried out on a few models chosen by its authors, and several studies release no code, so applying an analysis to another model or comparing two analyses on the same model requires reimplementing them.

1.1 What do existing studies examine?

A central question is whether different experts learn meaningfully different skills. For example, does one expert specialize in a particular domain while another specializes in a different one? Recent studies have examined this question from different perspectives and using different definitions of specialization.

Do et al. (2026) identify domain-specific experts and steer them to improve accuracy without retraining, Wang et al. (2026b) find a small domain-invariant committee of experts that receives most of the routing mass in every domain, and Wang et al. (2026a) show that similarity in expert usage follows similarity in hidden states rather than domain.

Herbst et al. (2026) report that experts act as fine-grained task specialists whose neurons are less polysemantic than those of dense feed-forward layers, and Lasy et al. (2026) explain routing decisions with sparse autoencoder features. Other studies decode the output of individual experts in vocabulary space (Chaudhari et al., 2025), compare expert parameters and outputs (Lo et al., 2025), relate routing to the local geometry of expert functions (Liu, 2026), and study routing in controlled or non-language settings (Fan et al., 2024; Ternovtsii & Bilak, 2026b;a; Chaudhari et al., 2026; Tangtartharakul & Storrs, 2026).

1.2 Why moe-interp?

Even simple questions can be measured in different ways. For example, if we ask “How important is expert 5 for biology questions?”, we could look at how often the expert is selected, how much routing weight it receives, or how much its output affects the model. Different studies use different measures for this (Do et al., 2026; Wang et al., 2026b; Chaudhari et al., 2025), so their results are not always directly comparable.

Different analyses also need access to different parts of the model, such as routing scores, expert activations, parameters, or gradients. These signals are not always available in the same form across different MoE architectures. As a result, applying an existing analysis to a new model often requires changing both the implementation and the way the model’s routing is handled.

With moe-interp, we aim to provide these signals through a common interface, clearly define the quantities used by each analysis, and state when an analysis cannot be applied to a particular model. This makes it easier to run and compare the same analyses across different MoE models.

2 moe-interp Models

We plan to support the models listed in Table 1. Table 2 lists the studies whose analyses the library will implement.

Model	N/k	Shared Experts	Full Scores	Studies
Mixtral 8x7B (Jiang et al., 2024)	8/2	✗	✓	2
DeepSeekMoE 16B (Dai et al., 2024)	64/6	✓	✓	3
Qwen1.5-MoE-A2.7B	60/4	✓	✓	2
DeepSeek-V2-Lite (DeepSeek-AI, 2024a)	64/6	✓	✓	2
OLMoE-1B-7B (Muennighoff et al., 2025)	64/8	✗	✓	5
DeepSeek-V3 (DeepSeek-AI, 2024b)	256/8	✓	✓	0
Qwen3-30B-A3B (Yang et al., 2025)	128/8	✗	✓	4
ERNIE-4.5-21B-A3B	64/6	✓	✓	3
gpt-oss-20b (OpenAI, 2025)	32/4	✗	✗	4
Geometric MoE (Ternovtsii & Bilak, 2026a)	1024/4	✗	✓	2

Table 1: Models we plan to support in moe-interp (N/k : routed experts and experts selected per token; Full Scores: a score exists for every expert before selection; Studies: number of the studies in Table 2 that analyse the model).

Study	Public Models	Public Code
Domain-specific experts (Do et al., 2026)	✓	✓
Closer look at MoE (Lo et al., 2025)	✓	✓
Geometric routing (Ternovtsii & Bilak, 2026b)	✗	✗
CommitteeAudit (Wang et al., 2026b)	✓	✓
MoE Lens (Chaudhari et al., 2025)	✓	✓
The Expert Strikes Back (Herbst et al., 2026)	✓	✓
RouterInterp (Lasy et al., 2026)	✓	✓
Equifinality (Ternovtsii & Bilak, 2026a)	✗	✗
Myth of specialization (Wang et al., 2026a)	✓	✗
Sparse crosscoders (Chaudhari et al., 2026)	✗	✗
MoE design choices (Fan et al., 2024)	✗	✗
Spectral geometry (Liu, 2026)	✗	✓
Vision expert tuning (Tangtartharakul & Storrs, 2026)	✗	✗

Table 2: Studies we plan to implement in moe-interp (Public Models: the study analyses publicly released pretrained models; Public Code: the authors released code).

Prerequisites. You do not need prior experience with MoE interpretability. Familiarity with Python, PyTorch, and basic neural-network or transformer concepts is enough to get started. We’ll help you learn the required MoE and interpretability concepts through the course of the project.

If this project interests you, please fill in the form below by **18 October, 2026, 11:59 PM IST**:

<https://forms.gle/5aqgjDowL3RgpsDf9>

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